Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	4 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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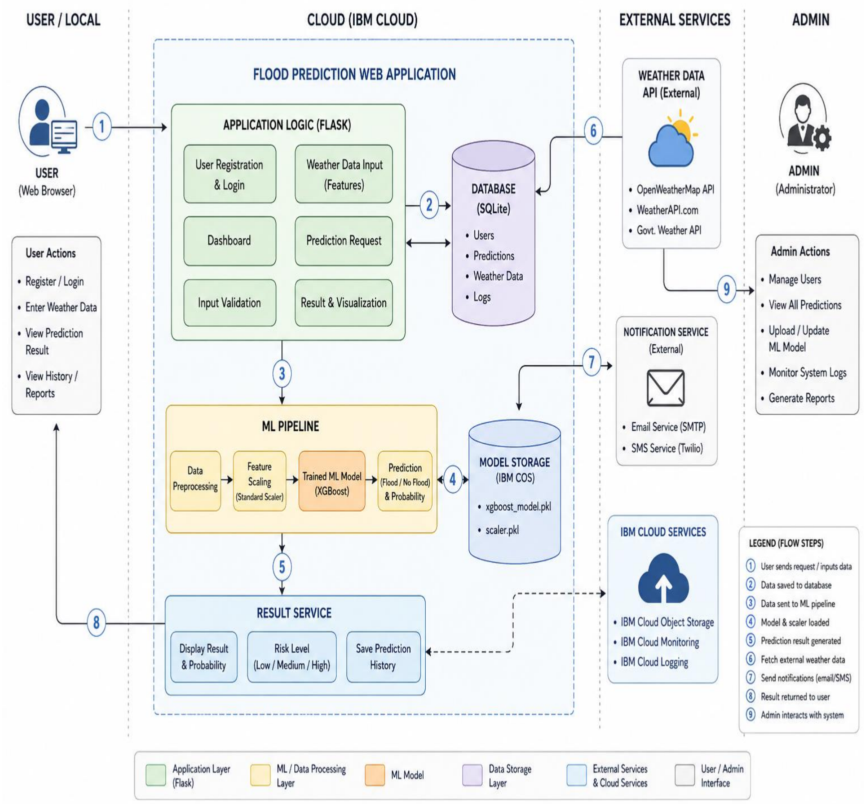
Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for users to register, log in, enter weather parameters, and view flood prediction results.	HTML, CSS, JavaScript, Bootstrap
2.	Application Logic-1	Handles user authentication, routing, dashboard, and request processing.	Flask (Python)
3.	Application Logic-2	Validates user input, preprocesses weather data, and performs feature scaling before prediction.	Python, Scikit-learn (StandardScaler), Joblib
4.	Application Logic-3	Loads the trained machine learning model and generates flood predictions.	XGBoost, Scikit-learn
5.	Database	Stores user information, weather inputs, prediction history, and system logs.	SQLite
6.	Cloud Database	Stores prediction records and application data after cloud deployment (optional).	IBM Cloud Databases / IBM Db2
7.	File Storage	Stores trained ML model, scaler object, datasets, and generated reports.	IBM Cloud Object Storage / Local File System
8.	External API-1	Retrieves live weather information for prediction (optional enhancement).	OpenWeatherMap API / WeatherAPI
9.	External API-2	Sends alerts or notifications to users (optional enhancement).	Twilio API / SMTP Email API
10.	Machine Learning Model	Predicts flood occurrence using historical weather data and selected features.	XGBoost, Decision Tree, Random Forest, KNN
11.	Infrastructure (Server / Cloud)	Hosts the Flask application and ML model for public access. :	IBM Cloud, Docker (optional), Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks and libraries used to develop the web application and machine learning model.	Flask, Python, Scikit-learn, XGBoost, Pandas, NumPy
2.	Security Implementations	Secure user authentication, input validation, password protection, and secure communication.	Flask Authentication, Werkzeug Password Hashing, HTTPS
3.	Scalable Architecture	Modular architecture separating UI, application logic, machine learning, and database for easy scalability.	Flask, IBM Cloud, REST API
4.	Availability	The application is deployed on the cloud and can be accessed by authorized users anytime.	IBM Cloud, IBM Cloud Object Storage
5.	Performance	Optimized preprocessing and prediction pipeline to provide fast and accurate flood predictions with minimal response time.	XGBoost, Joblib, Scikit-learn, SQLite

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d